

Cross-Technology Network Performance Analysis: 4G, 5G, and WiFi Mesh

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Simulation Approach

Simulations were conducted using NS-3 network simulator across three technologies (WiFi Mesh, 4G LTE, and 5G NR) following a systematic matrix approach. **STA/UE nodes were positioned using uniform random distribution** across the 400×400×30 m stage, avoiding building obstacles. A comprehensive scenario matrix was generated covering all combinations of: 3 RNG seeds (6, 7, 8), 3 node counts (5, 10, 15), and 3 payload sizes (10KB, 50KB, 1MB), with WiFi Mesh additionally varying across 2 frequency bands (2.4 GHz and 5 GHz).

Each simulation executed for 30 seconds with Gauss-Markov 3D mobility model for STA/UE movement. Network performance metrics (PDR, Throughput, Delay, Jitter) were collected using FlowMonitor and aggregated across seeds to compute mean values, standard deviations, and statistical distributions. Results were analyzed to compare technology performance across scalability (node count variation), traffic load impact (payload size variation), and overall reliability metrics.

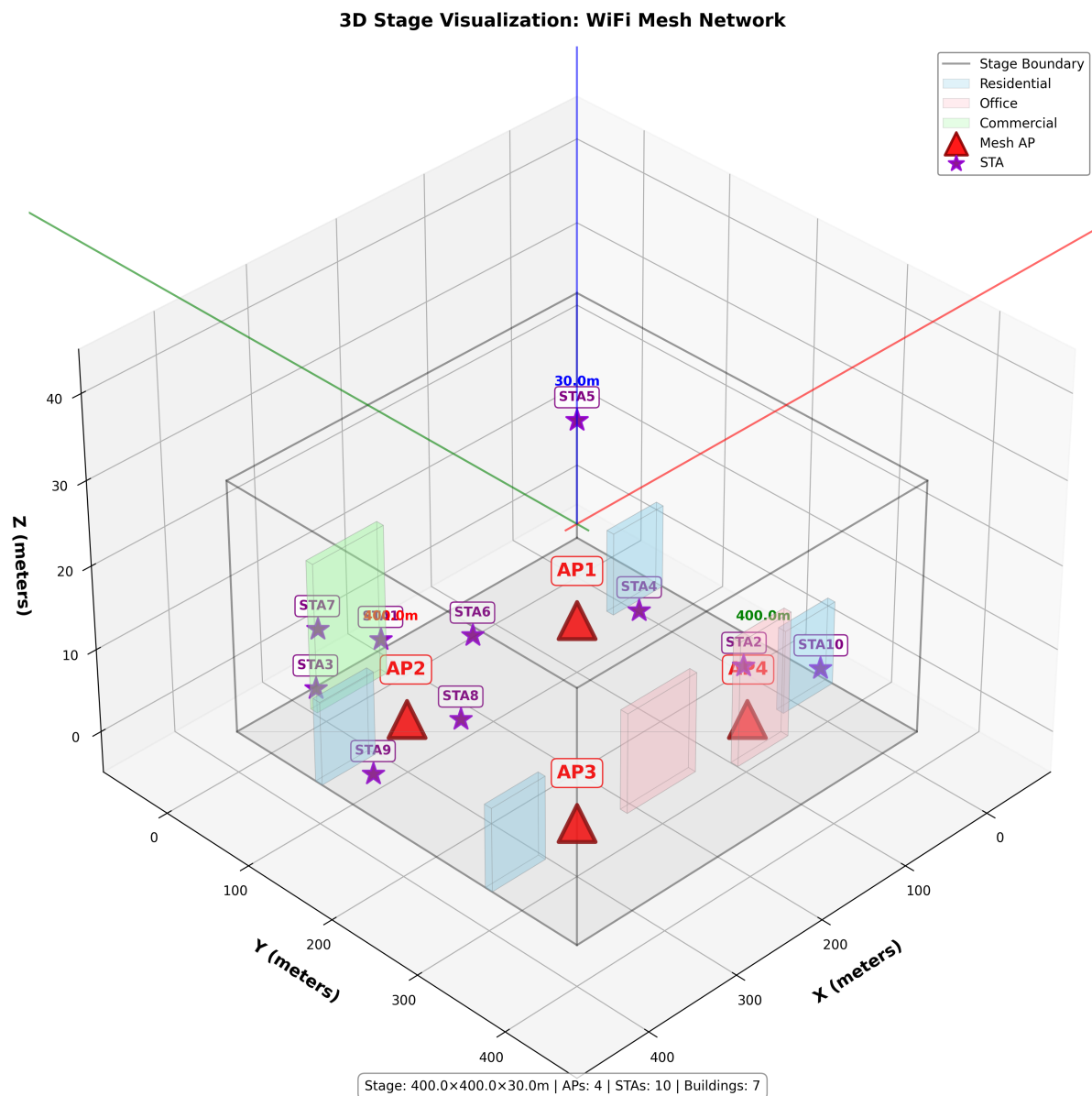
- Scenarios analyzed: 108

Common Configuration

Metric	WiFi Mesh	4G	5G
Stage Structure: 400×400×30 m	✓	✓	✓
Building Obstacles: 7 buildings	✓	✓	✓

Metric	WiFi Mesh	4G	5G
Propagation Loss Model: HybridBuildingsPropagationLossModel	✓	✓	✓
RNG Seeds: 6, 7, 8	✓	✓	✓
Simulation Time: 30.0 s	✓	✓	✓
Node Counts: 5, 10, 15	✓	✓	✓
Payload Size: 10KB, 50KB, 1MB	✓	✓	✓
UE/STA Mobility: Gauss-Markov 3D movement	✓	✓	✓
UE/STA TX Power: 15.0 dBm	✓	✓	✓
Traffic Flows: HTTP, HTTPS, Video (TCP), VoIP (UDP), Mixed	✓	✓	✓

WiFi Mesh Network Visualization



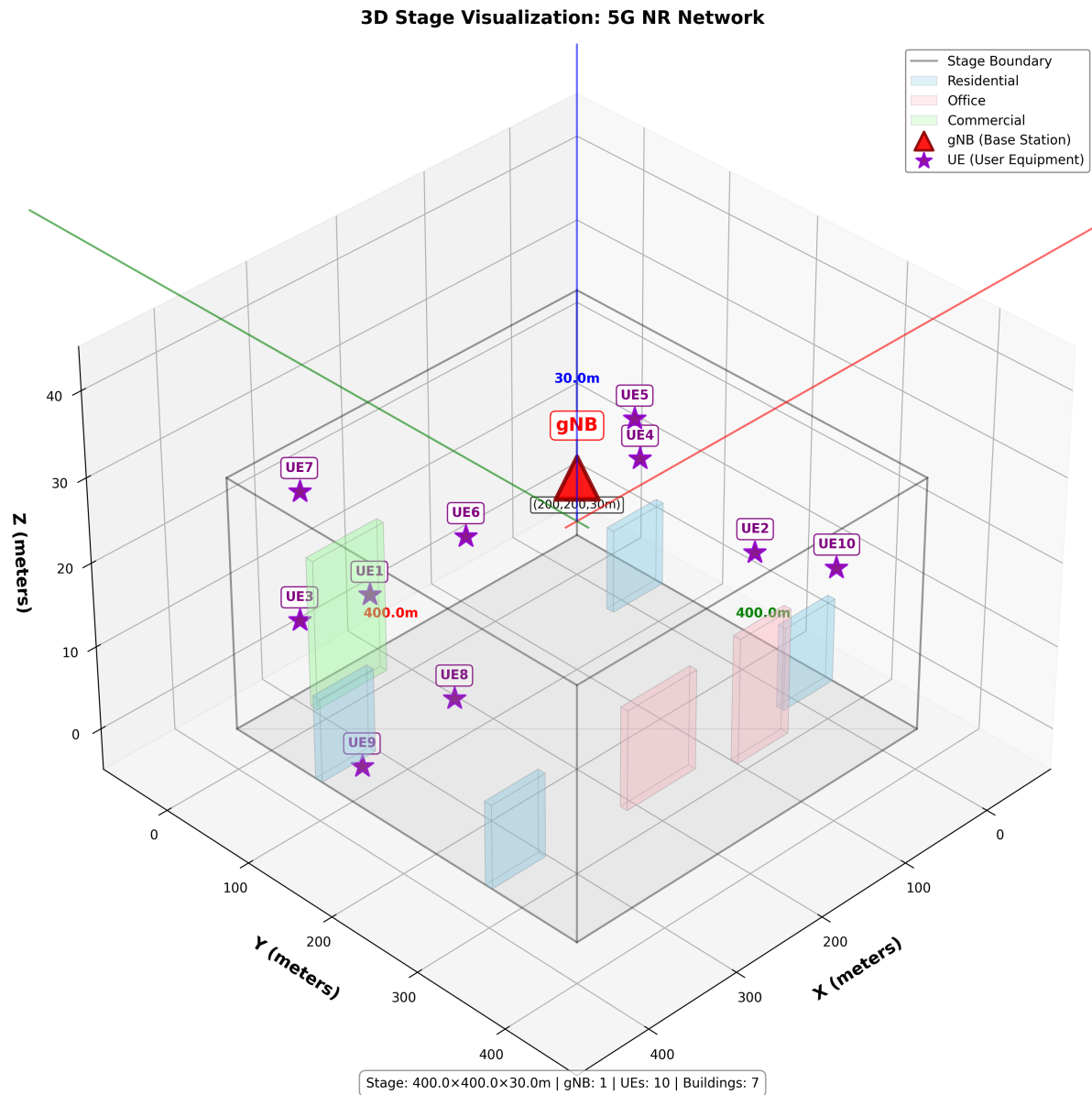
WiFi Mesh Configuration

Parameter	Value
Mesh Protocol	802.11s (Dot11sStack)
WiFi Standard (Mesh Backhaul)	802.11n @ 2.4 GHz
Data Mode	HtMcs7

Parameter	Value
TX Power (Mesh Backhaul)	23.0 dBm
TX Power (Hotspot/AP)	22.0 dBm @ 5 GHz
RX Sensitivity	-90.0 dBm
Antenna Gain (RX/TX)	5.0 dB / 5.0 dB
Number of Mesh Interfaces	1
Hotspot Band	5 GHz (802.11ac)

Reference: Values sourced from the TP-Link EAP225-Outdoor specifications at <https://www.tp-link.com/us/business-networking/omada-sdn-access-point/eap225-outdoor/v1/#specifications>.

5G Network Visualization



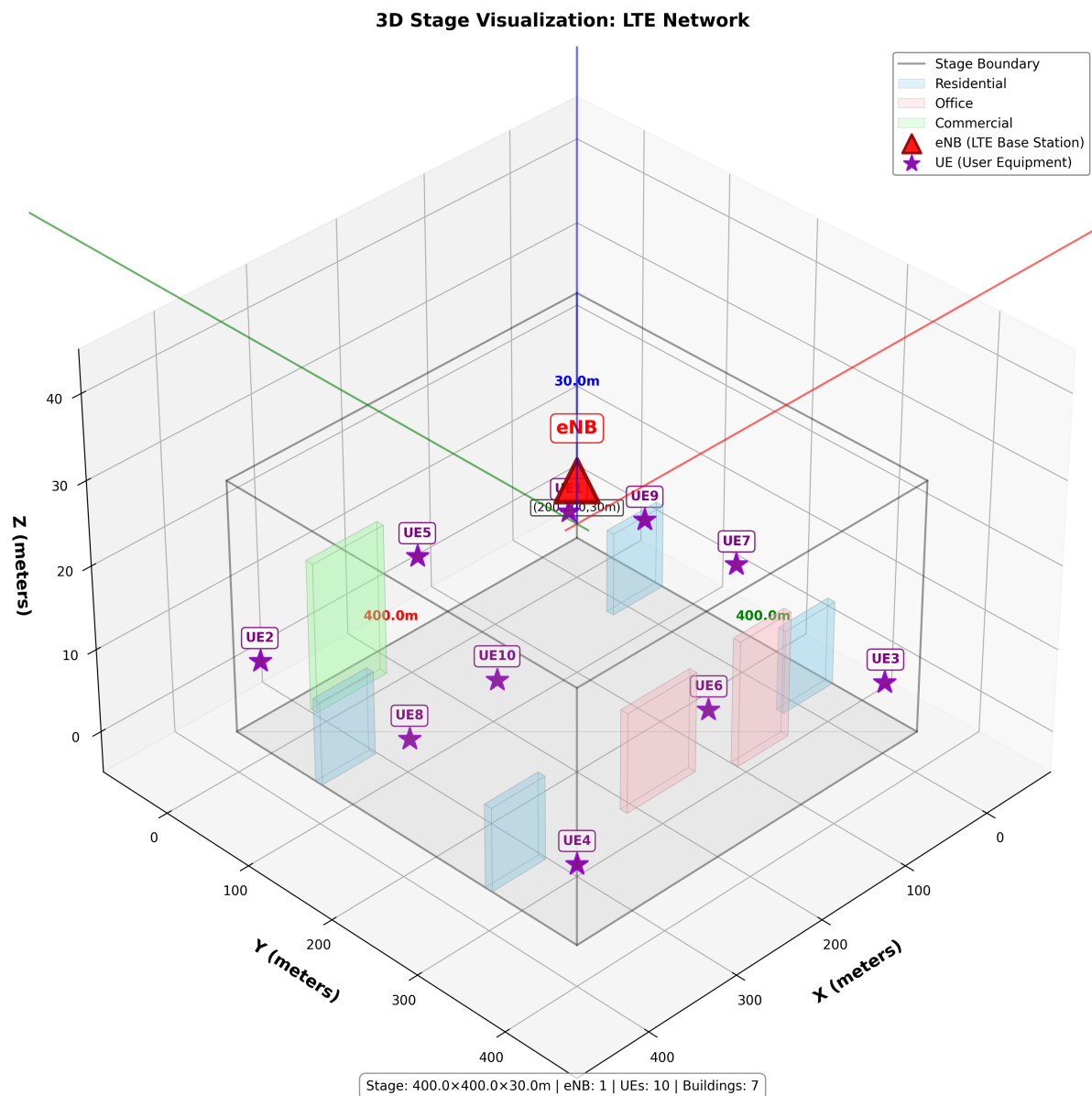
5G NR Configuration

Parameter	Value
Central Frequency	3.5 GHz
Bandwidth	100 MHz
Channel Model	RMa (Rural Macro)

Parameter	Value
Channel Model Type	ThreeGpp
Beamforming Algorithm	IdealBeamformingHelper (DirectPathBeamforming)
gNB TX Power	~43 dBm (macro cell)
Number of Component Carriers	1
Number of Bandwidth Parts	1

Reference: Values derived from the NS-3 default NR configuration.

4G Network Visualization



4G LTE Configuration

Parameter	Value
Frequency	2.0 GHz
eNB TX Power	43.0 dBm
Handover Algorithm	A3RsrpHandoverAlgorithm

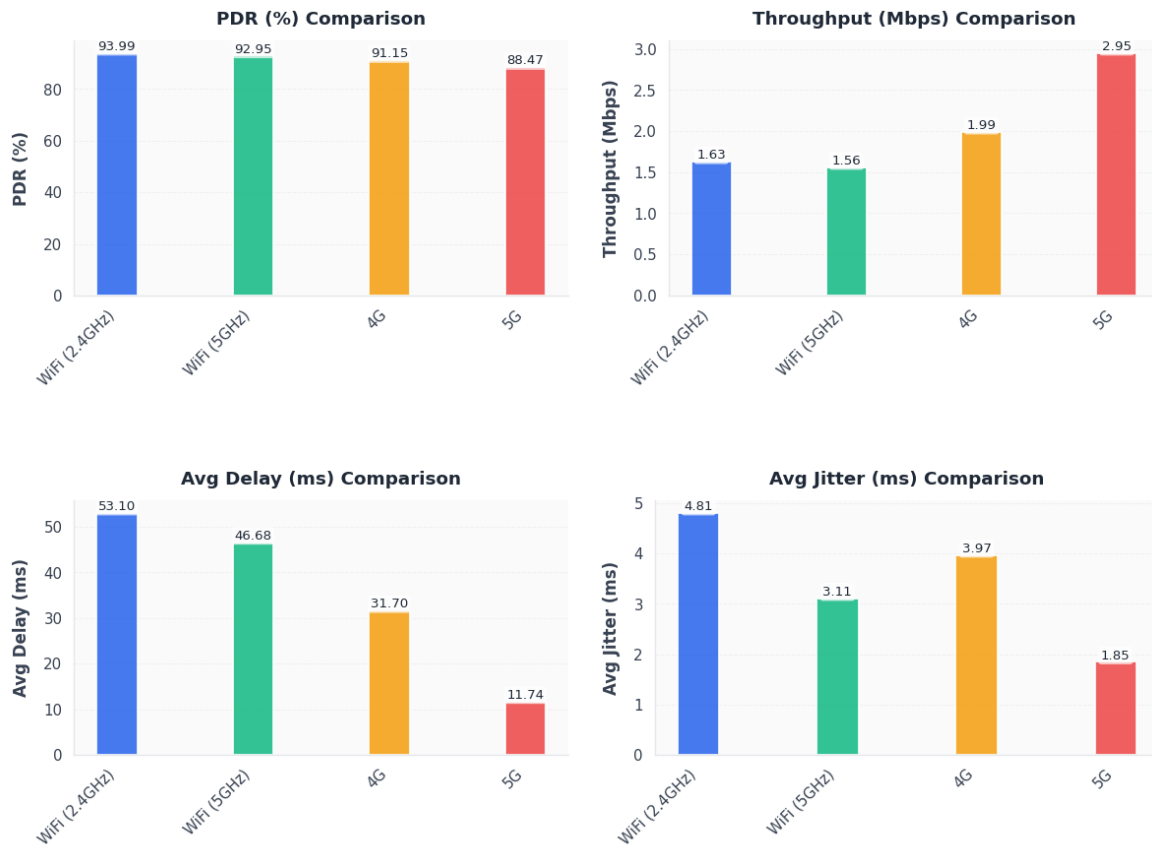
Parameter	Value
Handover Hysteresis	3.0 dB
Time To Trigger	160 ms

Reference: Values derived from the NS-3 default LTE configuration.

1. Cross-Technology Performance

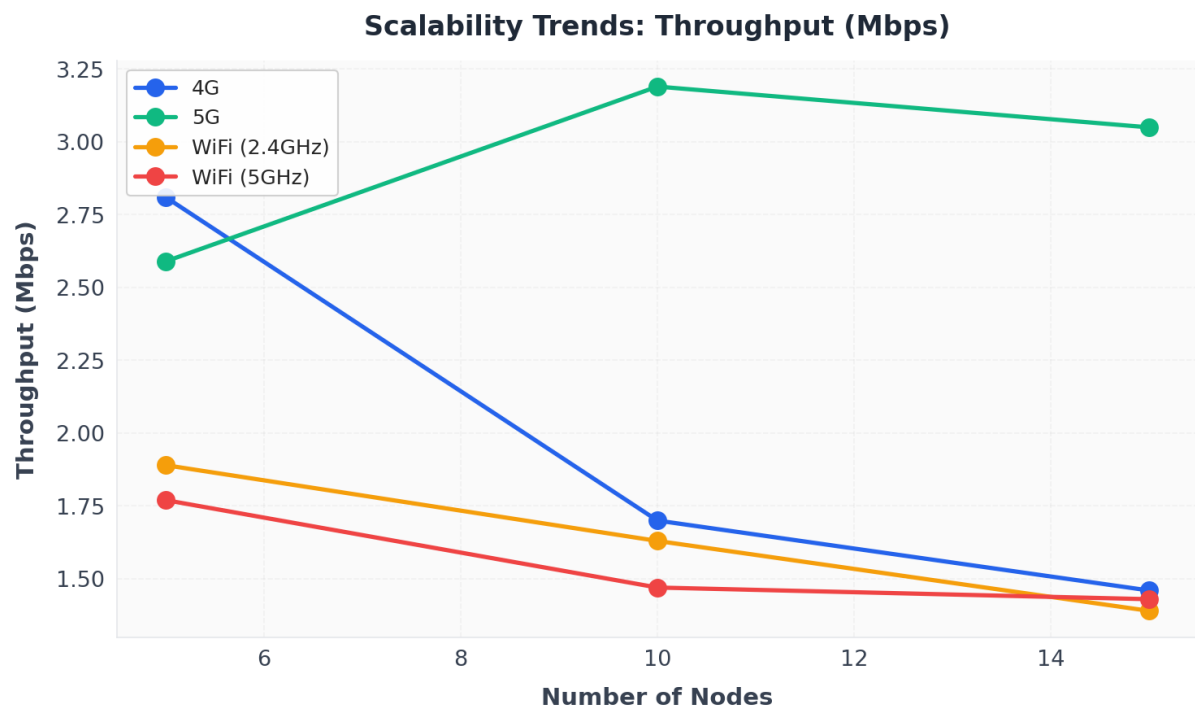
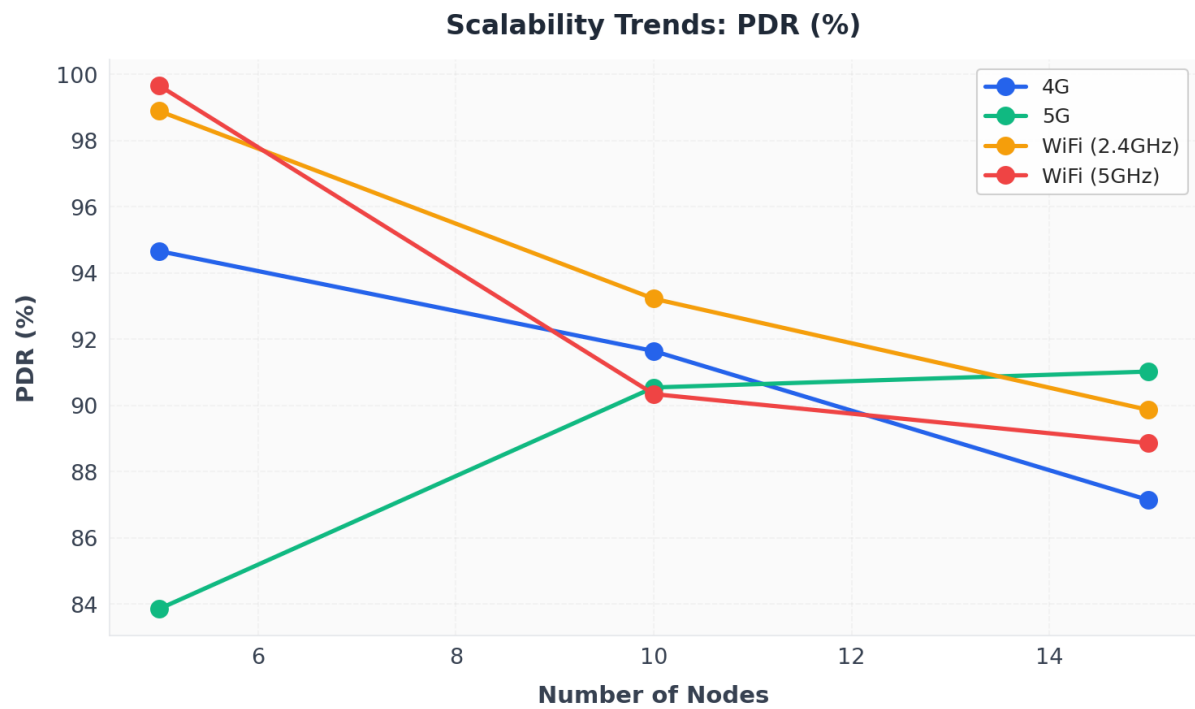
- 5G achieves the highest Throughput (Mbps) (2.95).
- WIFI Mesh (2.4 GHz) achieves the highest PDR (%) (93.99).
- 5G achieves the lowest Avg Delay (ms) (11.74).

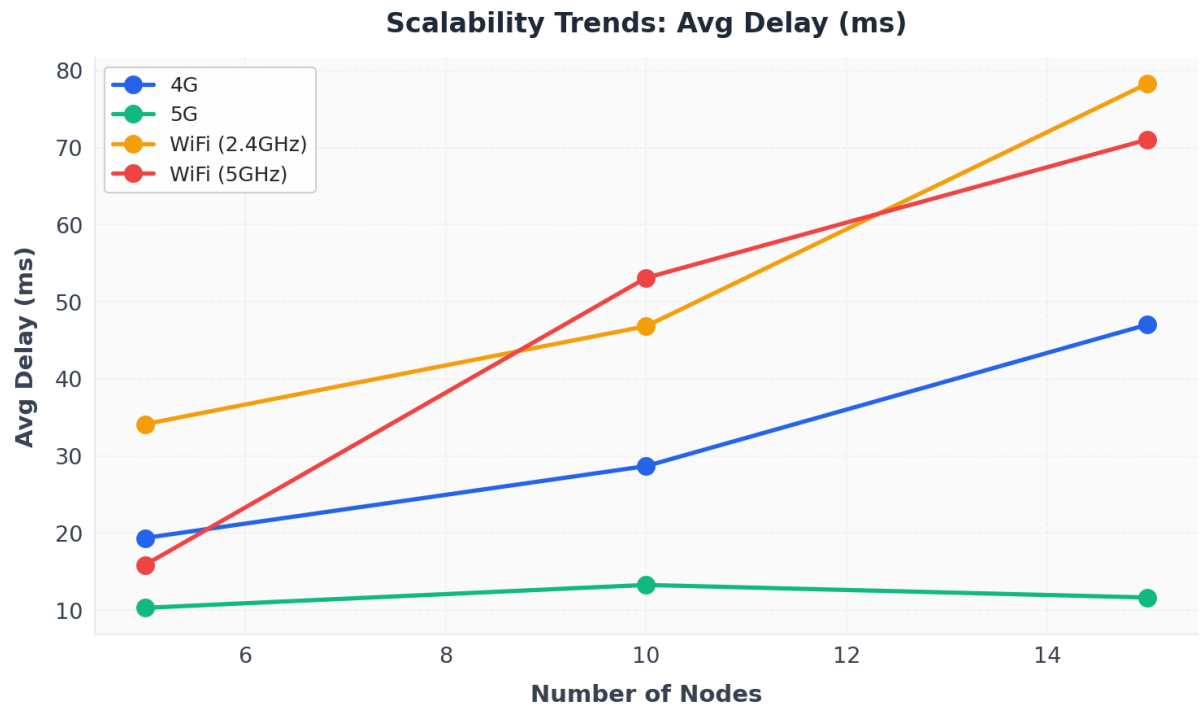
Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)	Mean Jitter (ms)
WIFI Mesh (2.4 GHz)	93.99	1.63	53.10	4.81
WIFI Mesh (5 GHz)	92.95	1.56	46.68	3.11
4G	91.15	1.99	31.70	3.97
5G	88.47	2.95	11.74	1.85



2. Scalability Trends

- WiFi Mesh (2.4 GHz) loses 9.04 percentage points of PDR and adds 44.18 ms delay between 5 and 15 nodes.
- WiFi Mesh (5 GHz) loses 10.79 percentage points of PDR and adds 55.19 ms delay between 5 and 15 nodes.
- 4G loses 7.52 percentage points of PDR and adds 27.67 ms delay between 5 and 15 nodes.
- 5G gains 7.17 percentage points of PDR and adds 1.34 ms delay between 5 and 15 nodes.





5 Nodes

Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)
WIFI Mesh (2.4 GHz)	98.89	1.89	34.13
WIFI Mesh (5 GHz)	99.66	1.77	15.86
4G	94.66	2.81	19.36
5G	83.85	2.59	10.30

10 Nodes

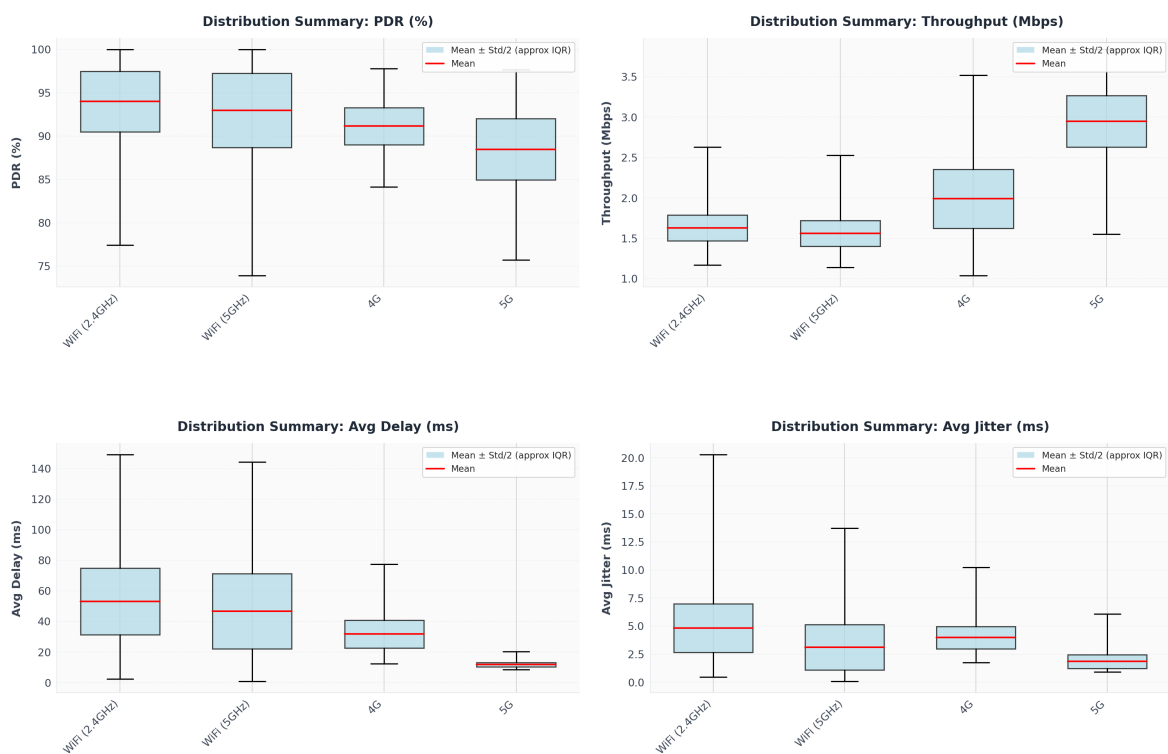
Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)
WIFI Mesh (2.4 GHz)	93.22	1.63	46.84
WIFI Mesh (5 GHz)	90.34	1.47	53.11
4G	91.64	1.70	28.69
5G	90.54	3.19	13.27

15 Nodes

Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)
WIFI Mesh (2.4 GHz)	89.86	1.39	78.31
WIFI Mesh (5 GHz)	88.86	1.43	71.05
4G	87.14	1.46	47.04
5G	91.02	3.05	11.64

3. Statistical Summary

- Tables below summarize distribution of core KPIs across seeds, node counts, and flow scales.



PDR (%)

Technology	Mean	Std Dev	Min	Max
WIFI Mesh (2.4 GHz)	93.99	6.99	77.41	100
WIFI Mesh (5 GHz)	92.95	8.56	73.92	100
4G	91.15	4.26	84.15	97.77
5G	88.47	7.08	75.72	97.66

Throughput (Mbps)

Technology	Mean	Std Dev	Min	Max
WIFI Mesh (2.4 GHz)	1.63	0.32	1.17	2.63
WIFI Mesh (5 GHz)	1.56	0.32	1.14	2.53
4G	1.99	0.73	1.04	3.52
5G	2.95	0.64	1.55	3.84

Avg Delay (ms)

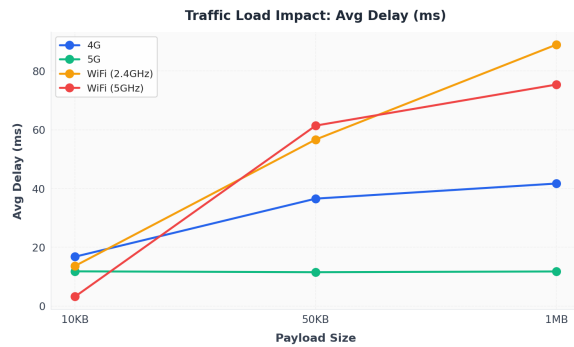
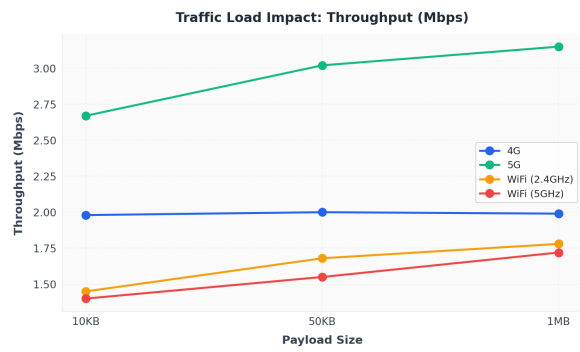
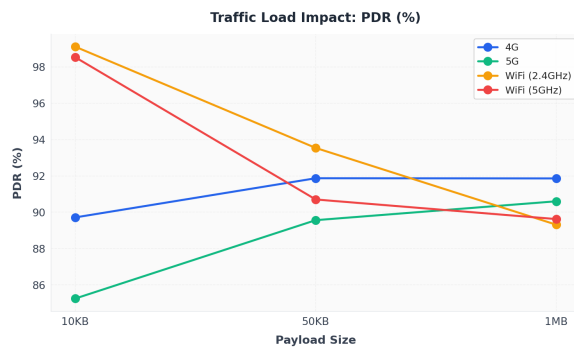
Technology	Mean	Std Dev	Min	Max
WIFI Mesh (2.4 GHz)	53.10	43.39	2.31	149.15
WIFI Mesh (5 GHz)	46.68	49.11	0.90	144.11
4G	31.70	18.29	12.32	77.35
5G	11.74	2.96	8.53	20.21

Avg Jitter (ms)

Technology	Mean	Std Dev	Min	Max
WIFI Mesh (2.4 GHz)	4.81	4.33	0.45	20.29
WIFI Mesh (5 GHz)	3.11	4.06	0.09	13.73
4G	3.97	1.96	1.74	10.25
5G	1.85	1.22	0.91	6.08

4. Traffic Load Impact

- WIFI Mesh (2.4 GHz) loses 9.81 percentage points of PDR and adds 75.22 ms delay from 10KB to 1MB payload.
- WIFI Mesh (5 GHz) loses 8.92 percentage points of PDR and adds 72.08 ms delay from 10KB to 1MB payload.
- 4G gains 2.15 percentage points of PDR and adds 24.87 ms delay from 10KB to 1MB payload.
- 5G gains 5.36 percentage points of PDR and reduces 0.04 ms delay from 10KB to 1MB payload.



10KB Payload

Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)
WIFI Mesh (2.4 GHz)	99.12	1.45	13.70
WIFI Mesh (5 GHz)	98.54	1.40	3.27
4G	89.71	1.98	16.83
5G	85.24	2.67	11.85

50KB Payload

Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)
WIFI Mesh (2.4 GHz)	93.54	1.68	56.67
WIFI Mesh (5 GHz)	90.70	1.55	61.40
4G	91.87	2.00	36.57

Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)
5G	89.56	3.02	11.56

1MB Payload

Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)
WIFI Mesh (2.4 GHz)	89.31	1.78	88.92
WIFI Mesh (5 GHz)	89.62	1.72	75.35
4G	91.86	1.99	41.70
5G	90.60	3.15	11.81

Summary

This analysis of 108 simulation scenarios demonstrates that **WiFi Mesh performs well** and offers a **cost-effective alternative** to 4G and 5G technologies.

WiFi Mesh (2.4 GHz) achieves the highest Packet Delivery Ratio (PDR) of 93.99%, indicating superior reliability. While 5G leads in throughput (2.95 Mbps) and latency (11.74 ms), WiFi Mesh maintains competitive performance with 1.63 Mbps throughput and shows robust scalability across different node counts.

WiFi Mesh is highly cost-effective compared to cellular technologies. It uses unlicensed spectrum (2.4 GHz and 5 GHz), requires no spectrum licensing fees, and can be deployed with standard commercial equipment. The mesh protocol (802.11s) enables self-organizing networks, reducing deployment complexity and operational costs.

Conclusion: WiFi Mesh is **feasible and cost-effective for implementation** in scenarios prioritizing reliability and cost-efficiency. Its strong PDR performance, combined with significantly lower deployment and operational costs compared to 4G and 5G, makes it a compelling choice for many network deployment scenarios.